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Research

At what spatial scale(s) do mammals respond to urbanization?

EDITOR'S CHOICE

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Spatial scale is fundamental in understanding species–landscape relationships because species' responses to landscape characteristics typically vary across scales. Nonetheless, such scales are often unidentified or unreliably predicted by theory. Many landscapes worldwide are urbanizing, yet the spatial scaling of species' responses to urbanization is poorly understood. We investigated the spatial scaling of urbanization effects on a community of 15 mammal species using ~60 000 wildlife detections collected from a constellation of 207 camera traps across an extensive urban park system. We embedded a bivariate Gaussian kernel in hierarchical multi-species models to determine two scales of effect (a scale of maximal effect and a broader scale of cumulative landscape effect) for two biological responses (occupancy and site visit frequency) across two seasons (winter and summer) for each species. We then assessed whether scales of effect varied according to theoretical predictions associated with biological responses and species traits (body size and mobility). Scales of effect ranged from < 50 m to > 9000 m and varied among species, but not as predicted by theory. Species' occupancy generally showed a weak response to urbanization and the scale of this effect was both highly uncertain and consistent across species. We did not detect any relationship between scales of effect and species' body size or mobility, nor was there any evident pattern of scaling across biological response or seasons. These results imply that 1) urbanization effects on mammals manifest across a very broad spectrum of spatial scales, and 2) current theories that a priori predict the scale at which urbanization affects mammals may be of limited use within a given system. Overall, this study suggests that developing general theory regarding the scaling of species–landscape relationships requires additional empirical work conducted across multiple species, systems and timescales.

Keywords: characteristic scale of effect, occupancy, response grain, spatial scaling, species–landscape relationships

Introduction

Scale is a foundational concept in ecology (Weins 1989, Levin 1992, Schneider 2001, Turner and Gardner 2015). Many ecological processes, from animal dispersal

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to community assemblage, result in characteristic patterns of variation across particular temporal and spatial scales (Mayor et al. 2009, Chave 2013). Investigating such patterns across scales to discover underlying ecological processes is a central aim of landscape and spatial ecology (Turner and Gardner 2015). Although the idea of linking observed patterns across a landscape to the processes that give rise to them has been widespread for decades (Allen and Hoekstra 1992, Turner and Gardner 2015), ecosystem complexity makes finding such linkages difficult (Allen and Starr 1982, Miguet et al. 2016, Boyce et al. 2017).

Spatial scale is crucial when examining species–landscape relationships. The effect of a landscape variable on a given species and for a given biological response is typically strongest at a particular spatial scale (Holland et al. 2004, Aue et al. 2012). This scale has been referred to by various terms, including the functional or characteristic scale of effect (Holland and Yang 2016). Here, we use the phrase scale of effect (following Jackson and Fahrig 2012, Moraga et al. 2019). Mismatch between the scale of analysis (defined by the researcher) and the scale of effect (defined by the true relationship) represents a version of the Modifiable Areal Unit Problem (Gehlke and Biehl 1934), whereby inference becomes an artifact of spatial sampling scheme rather than an ecological process (Jelinski and Wu 1996). In modeling species–landscape relationships, this mismatch can lead to underestimation of variation, reduced predictive accuracy, biased magnitudes and incorrect directionality of effects, and imprecise estimation of the relative importance of landscape variables (Holland et al. 2004, De Knegt et al. 2010, 2011, Smith et al. 2011, Miguet et al. 2016, Mertes and Jetz 2018). Accurately identifying the scale of effect can aid in theory-building regarding species–landscape relationships (Weins 1989, Levin 1992) and species' conservation efforts by identifying the scales at which landscapes should be managed (Miguet et al. 2017, Gallo et al. 2018).

Several theoretical predictions have been made regarding scales of effect for species–landscape relationships (reviewed by Miguet et al. 2016). Scale of effect is expected to vary according to species traits, with a positive relationship expected with both body size and mobility (O'Neill et al. 1988, Holland et al. 2005, Jackson and Fahrig 2012, Thornton and Fletcher 2014). This expectation stems from the observation that larger-bodied and more mobile species tend to use landscapes at broader scales to meet energetic and life history requirements (Holland et al. 2005, Thornton and Fletcher 2014). Scale of effect is also predicted to vary by biological response, with increasing scales expected when moving from local processes (e.g. fecundity and abundance) to those which occur across the broader landscape (e.g. occupancy and genetic diversity; Jokimäki and Huhta 1996, Jackson and Fahrig 2014, Martin 2018).

Despite the importance of scale in understanding species–landscape relationships, and the intuitive appeal of the theoretical predictions above, scales of effect remain poorly identified for many species and theory has often failed to align with empirical results (Wheatley and Johnson 2009, Jackson and Fahrig 2015, Boyce et al. 2017). For example, a global review of 71 studies that evaluated landscape variables' effects across multiple scales suggested that the true scale of effect was often more extreme than the scales analyzed – and thus unidentified (Jackson and Fahrig 2015). Another review of 23 studies that compared scales of effects across biological responses found that scales varied by the response measured, but not in the manner predicted by theory (Martin 2018). These findings suggest that scale of effect remains difficult to predict a priori and often goes unidentified in studies of species–landscape relationships, thereby potentially resulting in widespread bias, misspecification or misinterpretation of species' responses to landscape characteristics (Holland et al. 2004, Wheatley and Johnson 2009, De Knegt et al. 2011). Identifying the scale of effect in landscapes undergoing rapid alteration, such as urban areas, is important because effective conservation and management requires accurate appraisals of species' response to such changes.

Many landscapes are urbanizing worldwide, a trend that is leading to undesirable outcomes and novel dynamics among the ecological communities that inhabit urban areas (McKinney 2006, Warren et al. 2006, Bateman and Fleming 2012). For example, urbanization can lead to a shifted balance of exotic and native species (Riley et al. 2005), elevated rates of species' endangerment and extinction (Czech et al. 2000), and rapid evolution in urban-dwelling animal populations (Johnson and Munshi-South 2017). Despite these observations, comprehensive evaluations of the scales at which these effects occur are virtually absent (Pickett et al. 2016, Moll et al. 2019a). The lack of clarity regarding the scales at which urbanization effects take place impedes a more generalized understanding of urban ecology and limits management and conservation efforts because the scale at which to concentrate such efforts is often unknown (Borgström et al. 2006, Gallo et al. 2018, Moll et al. 2019a).

Here, we evaluated the spatial scaling of urbanization effects on a mammal community composed of 15 species inhabiting an extensive urban park system in Cleveland, Ohio, USA. We used a kernel-based smoother embedded in hierarchical multi-species occupancy models to determine whether distinct scales of effect existed, and if so, to quantify their magnitudes and uncertainty. We then tested theoretical predictions regarding the relationship between scales of effect and species' traits and biological response. Specifically, we tested the predictions that scales of effect would: 1) increase with species' body size, 2) increase with species' mobility and 3) be larger for a landscape-level response (occupancy) than for a more local response (site visit frequency). We also tested whether scale of effect varied across seasons. Our analysis provides insight into the scales at which mammals respond to urbanization, the degree to which theory predicts such scales of effect, and the importance of scale in informing conservation and management.

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Material and methods

Study system

We conducted research in the Cleveland Metroparks (hereafter Metroparks) in Cleveland, Ohio, USA (Fig. 1). The Metroparks contain 9469 ha of parkland distributed among 18 reservations that occur along a gradient of urbanization (Moll et al. 2018). The reservations vary from 24 to 1611 ha in size (mean = 459 ha, SD = 480 ha) and contain mixed hardwood forests dominated by oak *Quercus* sp., beech *Fagus grandifolia* and maple *Acer* sp. trees. Forests, wetlands, grasslands, open areas (e.g. mowed grass) and urban development (e.g. roads, parking lots, buildings) are the primary land cover types that are interspersed throughout the Metroparks (Fig. 1). Reservations near the city center tend to be more urbanized (i.e. surrounded by residential, commercial and industrial lands), while reservations farther from the city center are typically enveloped in a rural landscape within a matrix of large-lot single family homes (Fig. 1).

Surveying the mammal community

We surveyed the mammal community of the Metroparks across several seasons (i.e. winter, spring and summer; from 7 December 2015 to 30 September 2016). We established 207 study sites throughout the Metroparks using a Generalized Random Tessellation Stratified (GRTS) study design, which resulted in a spatially balanced sample within and among reservations (Stevens and Olsen 2003). In December 2015, we placed camera traps at 104 sites and in July 2016 we placed camera traps at an additional 103 sites. At each site, we affixed Bushnell Trophy Cam HD Aggressor camera traps to trees ~50 cm above the ground. We did not bait sites and employed a systematic protocol to choose each camera trap location (Supplementary material Appendix 1).

We examined the scaling of urbanization effects on 15 mammal species in the camera trap images. Overall, these species' body sizes and home ranges varied across several orders of magnitude (i.e. from 0.2 to 75 kg and from < 0.1 to 19 km^2, respectively; see below). The species included coyotes *Canis latrans*, domestic cats *Felis silvestris*, eastern chipmunks *Tamias striatus*, eastern cottontails *Sylvilagus floridanus*, eastern gray squirrels *Scuirus carolinensis*, fox squirrels *Sciurus niger*, long-tailed weasels *Mustela frenata*, mink *Neovison vison*, raccoons *Procyon lotor*, red foxes *Vulpes vulpes*, red squirrels *Tamiasciurus hudsonicus*, striped skunk *Mephitis mephitis*, Virginia opossums *Didelphis virginiana*, white-tailed deer *Odocoileus virginianus* and woodchucks *Marmota monax*. Given the difficulty in distinguishing between mink and multiple weasel species, particularly in night-time images and with respect to juvenile animals, we combined mink and weasel images into a single category according to their shared taxonomic family, Mustelidae.

Creating the urbanization covariate

There are many ways to quantify urbanization (reviewed by McIntyre et al. 2000, Moll et al. 2019a). Recent work has highlighted the utility of using multivariate techniques to

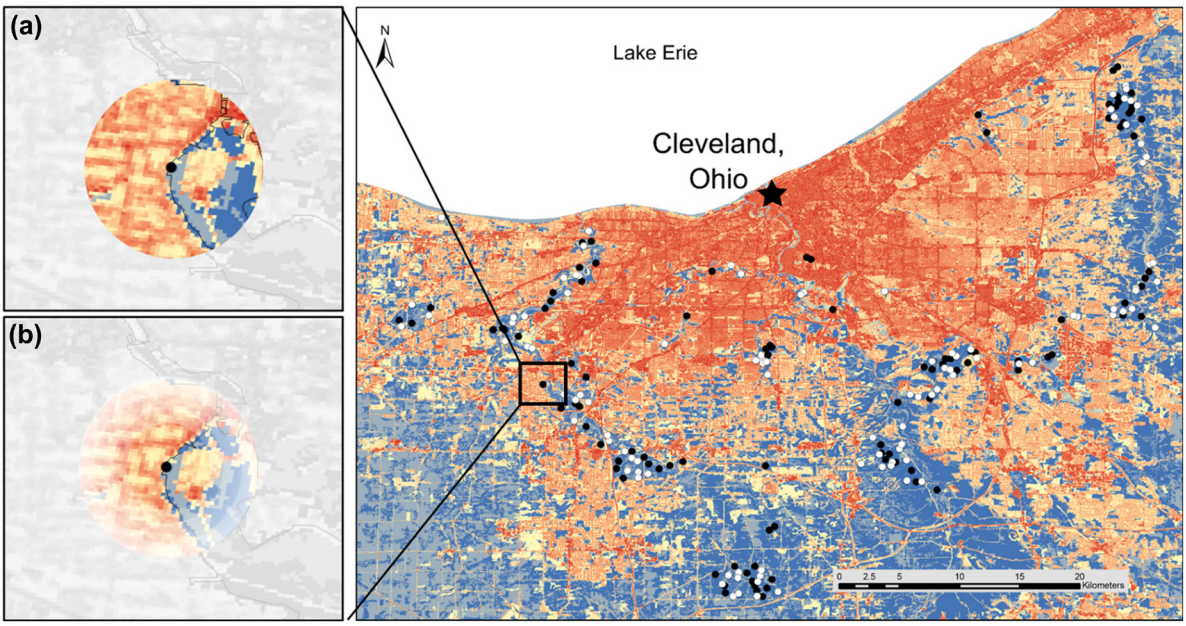

Figure 1. Locations of 207 camera traps deployed throughout the Cleveland Metroparks in Cleveland, Ohio, USA. The white circles indicate cameras deployed in December 2015 and black circles indicate cameras added in June 2016. The background cells depict urbanization, with red, yellow and blue pixels indicating high, medium and low urbanization, respectively (see text for details). Panel (a) shows a threshold-based approach to evaluating scale of effect, with urbanization quantified by calculating a mean within some buffer around a site. Panel (b) shows a weighted smoothing approach where urbanization values around a focal site are down-weighted with distance.

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create an omnibus urbanization axis for examining spatial patterns in ecological processes in urban areas (du Toit and Cilliers 2011, McDonnell et al. 2012, Moll et al. 2019a). This method was well-suited for pursuing our research objective, i.e. quantifying the scaling of mammal response to urbanization, broadly construed. Therefore, we used a principal components analysis (PCA) to quantify urbanization across our study area.

We conducted the PCA on three spatial representations of urbanization, each informed by 30 m resolution raster data from the 2016 National Land Cover Database (Homer et al. 2015). These included 1) a binary raster of developed and non-developed land use, 2) a binary raster of forested and non-forested land use and 3) a raster depicting the proportion impervious surface. As expected, these rasters were correlated, with pairwise Pearson correlations among them ranging from 0.4 to 0.7 across our study sites. We used the 'Rstoolbox' package (Leutner and Horning 2019) in the Program RStudio ver. 3.4.1 running R ver. 3.5.2 (RStudio Team 2015) to create a new 30 m resolution raster as the first principal component of these three rasters (depicted in Fig. 1). Given that this first component explained ~61% of the variance, we used it as the urbanization covariate in all models. The loadings of this component for the three variables listed above (i.e. the developed/non-developed, forested/non-forested and proportion impervious rasters) were 0.64, −0.32 and 0.70. Thus, large positive values of the urbanization covariate indicated developed locations with a high proportion of impervious surface and a lack of forest cover, whereas negative values indicated forested locations with little or no development or impervious surface.

Modeling wildlife occupancy and site visit frequency

We fit hierarchical multi-species occupancy models to evaluate the effect of urbanization on two biological responses: 1) species' occupancy probability at a given site, and 2) site visit frequency, given occupancy. Here, occupancy refers to a given species using a site at least once during a season and is a fundamentally important ecological parameter given its relevance to species' distributions, landscape use and metapopulation dynamics (Hanski 1998, MacKenzie et al. 2002, 2005, Tyre et al. 2003). Site visit frequency is a representation of species' intensity of use across the landscape, and thereby represents a finer-scale ecological process than occupancy with relevance to species' space use within their home range (Kays et al. 2016, Moll et al. 2018).

We modeled occupancy as a latent state variable $Z_{i,j}$ that equals 1 if species i occupies site j, and 0 otherwise (MacKenzie et al. 2002, Tyre et al. 2003). This variable accounted for imperfect detection (i.e. a species occupied a site but was not detected) and was Bernoulli distributed with probability $\Psi_{i,j}$, and we modeled the effects of covariates (i.e. urbanization and nuisance variables, see below) on $\Psi_{i,j}$ using a logit link function. We modeled site visit frequency as the number of times a given species visited a site per week, given occupancy (see Moll et al. 2018 for additional model details). This portion of the model took the form:

$$y_{i,j,k} \mid Z_{i,j} \sim \text{Poisson}\left(Z_{i,j} \times \rho_{i,j,k} \times \lambda_{i,j,k}\right)$$

where $y_{i,j,k}$ is the number of observed visits species i made to site j during week k, $\lambda_{i,j,k}$ is the expected value of the number of visits and $\rho_{i,j,k}$ is a random variable with a 1-parameter gamma distribution that accounts for overdispersion in the observation data (Greene 2008). We modeled the effects of covariates on expected species-specific visit frequencies ($\lambda_{i,j,k}$) using a log link function (Greene 2008). We only used data from cameras that were operational for the entirety of a given week. Thus, each weekly count had identical effort and no offset term was required in the model.

We fit winter/early spring and summer models, with winter/early spring delineated as December–April (hereafter winter for simplicity) and summer as June–September. We separated these seasons by the month of May because during that month the environmental conditions in the study area (e.g. temperature, precipitation) were variable and many species were transitioning from winter to summer activities (e.g. from breeding to young-rearing; Grund et al. 2002, Gosselink et al. 2007). Previous work has suggested that several nuisance variables might influence site visit frequencies and thus should be controlled for. These included a temperature covariate in the winter model (Moll et al. 2018), a covariate that quantified the degree to which vegetation obscured camera trap viewsheds in the summer model (Moll et al. unpubl.), and a distance to natural water sources covariate in both models (Hilty and Merenlender 2004). We created the distance to water covariate by calculating the Euclidean distance from each site to the nearest river, stream or pond. We did not include additional landscape covariates in the model because our primary objective was to evaluate the scaling of urbanization effects. Moreover, many other common land use variables (e.g. proportion forest cover) covary with urbanization, thereby precluding the usefulness of their inclusion in the same model (Cade 2015). Following the standard convention of multi-species occupancy modeling, we assumed that estimated species-level parameters were random effects governed by community-level hyper-distributions (Zipkin et al. 2009; see model code in Supplementary material Appendix 2).

Evaluating the scaling of urbanization response

A common method of determining scale of effect is to first calculate the mean of a landscape covariate in buffers of varying size around focal sites (Fig. 1a), then fit models using values from each buffer, and finally select a best model, and hence the scale of effect, using a metric such as AIC (Holland et al. 2004, Miguet et al. 2016, Martin 2018). This method, however, unrealistically assumes that the effect of a landscape covariate is constant within some buffer and then

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drops to zero beyond that distance (Aue et al. 2012; Fig. 1a). An alternative method employs a distance-weighted smoothing function (Fig. 1b). The smoothing function assumes that the effect of a landscape variable is strongest close to the focal site and decays as a function of distance (see the first column in Fig. 2), which provides a more realistic depiction of many ecological processes (e.g. movements of central place foragers; Orians and Pearson 1979). This approach can reduce model bias and lead to more accurate interpretations of species–landscape relationships (Aue et al. 2012, Chandler and Hepinstall-Cymerman 2016, Miguet et al. 2017).

To calculate a distance-weighted representation of the urbanization covariate (URB_w), we first discretized the landscape into non-overlapping concentric rings around each study site that were 50 m wide and had radii that increased from 50 to 10 000 m in increments of 50 m. We hypothesized that 10 000 m would be a sufficient maximum distance for capturing scales of effect, and this expectation was confirmed by the results in that none were suggestive that a larger scale of effect existed (Jackson and Fahrig 2015). We calculated URB_w at each of j sites by calculating a weighted average of the urbanization variable as:

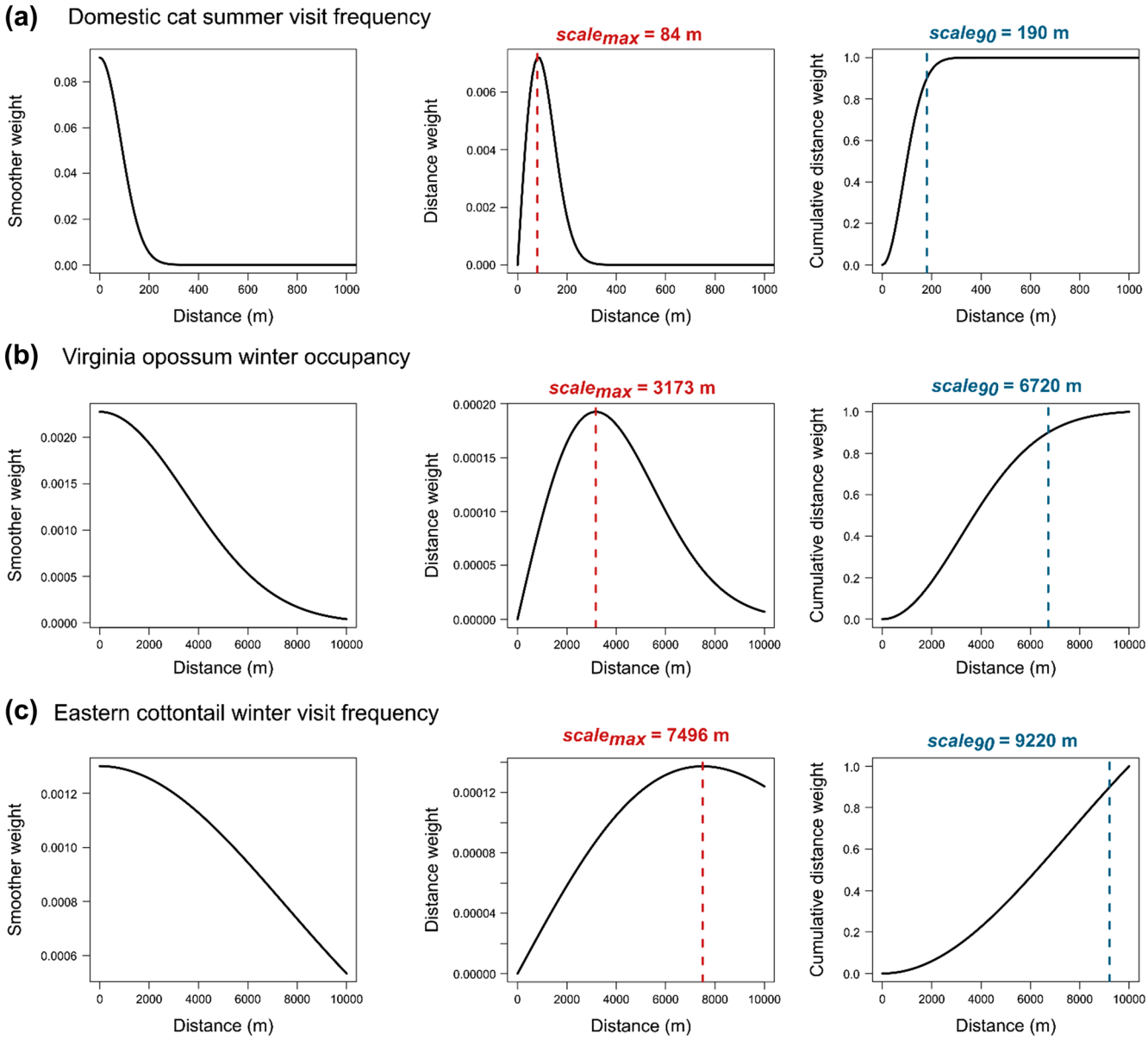

Figure 2. Examples showing the shape of a bivariate Gaussian smoothing kernel (first column), the distance weight function (middle column) and the cumulative weight function (third column) of an urbanization covariate for small (domestic cat summer visit frequency, panel a), medium (Virginia opossum winter occupancy, panel b) and large (eastern cottontail winter visit frequency) scales of effects. Two scales, the maximal effect ($scale_{max}$) and that which contains 90% of the smoother weight ($scale_{90}$), are depicted as dashed red and blue lines, respectively.

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$$\mathrm{URB}_{w(j)} = \sum_m w_m \times \mathrm{urb}_{m(j)}$$

where $\mathrm{urb}_{m(j)}$ is the mean value of the urbanization PCA raster described above in ring m encircling site j, and w_m is the weight of ring m calculated according to a Gaussian kernel:

$$w_m = \frac{\exp\left(-\frac{d^2_{m(j)}}{2\sigma^2}\right) \times A_m}{\sum_m \exp\left(-\frac{d^2_{m(j)}}{2\sigma^2}\right) \times A_m}$$

where $d_{m(j)}$ is the distance between site j and ring m (i.e. the ring's radius), A_m is the area of ring m, and σ is a scaling parameter estimated from the data that determines how rapidly the effect of urbanization decays with distance (Chandler and Hepinstall-Cymerman 2016, Yeiser et al. 2018). Note that w_m sums to one for each given site j. A large σ value produces a flat kernel such that locations far from the focal site have similar weights to those nearby (a large scale of effect). As noted above, theory predicts a larger σ for larger-bodied species or those exhibiting extensive landscape movements (e.g. long-distance dispersers; Jackson and Fahrig 2012). Conversely, a small σ value produces a kernel that sharply declines with distance, thereby assigning low weights to locations far from the focal site (a small scale of effect; Fig. 2). A small σ might be expected for smaller-bodied species and/or those with limited movement (O'Neill et al. 1988, Thornton and Fletcher 2014).

We used the equation above to calculate two scales of effect (following Miguet et al. 2017). First, we calculated the scale at which urbanization had a maximal effect on the response variable ($\mathrm{scale}_{\mathrm{max}}$), which is the distance at which w_m is maximal (Fig. 2, middle column; Aue et al. 2012). This scale is roughly equivalent to the scale at which a given covariate's coefficient is maximal in magnitude (Holland et al. 2004). Second, we determined the scale that contained 90% of the effect of urbanization (scale_{90}) by calculating the distance at which a circular area around the focal site contained 90% of w_m (Fig. 2, third column; Miguet et al. 2017). This scale is useful for situations where there is interest in more comprehensively capturing a landscape variable's effect on a given process (e.g. managing for endangered species habitat).

Evaluating theoretical expectations

We assessed whether scale of effect varied according to theoretical expectations related to species traits, biological responses and seasons. We examined the species traits of body size and mobility because these traits have been the focus of most theoretical predictions regarding the scaling of species–landscape effects (Roland and Taylor 1997, Thornton and Fletcher 2014, Miguet et al. 2016, Mertes and Jetz 2018). For biological responses, occupancy is expected to be associated with a larger average scale of effect than the more local process of site visit frequency (Miguet et al. 2016, Martin 2018). Finally, the scaling of landscape effects on wildlife could change seasonally. For example, many species alter their space use and movements during the young-rearing summer season (e.g. white-tailed deer; Grund et al. 2002), but it is generally unknown whether scales of effect vary synchronously with seasonal changes in species' behavior and movement.

To examine these potential relationships, we fit linear models of the log-transformed $\mathrm{scale}_{\mathrm{max}}$ as a function of 1) log-transformed body mass, 2) log-transformed home range size (following Thornton and Fletcher 2014), 3) biological response and 4) season. We obtained body mass and most home range values from the panTHERIA database (Jones et al. 2009). For several species, smaller home ranges have been observed in urban relative to non-urban populations. Thus, we used estimated urban home ranges for the following species: coyote (720 ha; Gehrt et al. 2010), raccoon (40 ha; Hadidan et al. 2010), red fox (60 ha; Soulsbury et al. 2010), striped skunk (126 ha; Gehrt 2004) and white-tailed deer (68 ha; Grund et al. 2002). In this analysis, we propagated uncertainty regarding $\mathrm{scale}_{\mathrm{max}}$ by fitting linear models in a Bayesian framework to every posterior estimate of $\mathrm{scale}_{\mathrm{max}}$ produced in the analysis described above ($N = 3000$; see below). We report the mean values of the posterior means and their 95% CIs. This two-step process accounts for uncertainty across both analysis steps (see Kery and Royle 2015, pp. 680–682; model code provided in Supplementary material Appendix 2).

Finally, in evaluating scales of effect, it is important to consider the intrinsic spatial structure of heterogeneity for a given landscape variable. Such variables typically exhibit spatial autocorrelation, where locations nearby are more similar than those more distant. The distance at which this spatial autocorrelation persists reflects the intrinsic scale of heterogeneity, and thus the typical patch size, for a landscape variable (Garrigues et al. 2006). If this intrinsic scale is fine (e.g. < 1000 m), then assessments of species–landscape relationships are especially vulnerable to bias if analyzed at an improper scale (Mertes and Jetz 2018). Additionally, if the intrinsic scale of heterogeneity differs from the scales of effect observed for species' responses, then the observed scales of effect represent non-random biological responses rather than mere reflections of predominant spatial landscape structure (Wheatley and Johnson 2009, De Knegt et al. 2011). We determined the intrinsic scale of heterogeneity for urbanization using a variogram, which quantifies spatial autocorrelation as a function of distance. The variogram's range (i.e. the distance at which it begins to asymptote, or reach its sill) indicates the intrinsic scale of heterogeneity (Garrigues et al. 2006). We used the R package 'usdm' (Naimi 2015) to calculate the range of the urbanization covariate by fitting an exponential model using non-linear least squares regression to an empirical variogram calculated from the urbanization raster (Wackernagel 2003, Mertes and Jetz 2018).

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Model analysis

We analyzed models using Monte Carlo Markov Chain (MCMC) simulations in a hierarchical Bayesian framework. For the scale of effect models, we ran three chains of 8000 iterations each following a burn-in of 2000 and thinned chains by eight (resulting in n = 3000 iterations). We used non-informative priors for all parameters (Zipkin et al. 2009). For the species trait, season and biological response models, we ran three chains of 1000 iterations each following a burn in of 200 and thinned chains by three. We evaluated model convergence by visually inspecting traceplots and ensuring all R-hat statistics were < 1.1 (Gelman and Hill 2007). We assessed the model fit by conducting posterior predictive checks and calculating Bayesian p-values, where values of $0.05 < p < 0.95$ indicate acceptable fit (Gelman et al. 2004, Kéry and Royle 2015). We evaluated the potential for spatial autocorrelation by plotting spline correlegrams of Pearson residuals (Rhodes et al. 2009, Zuckerberg et al. 2012). In parameter interpretation, we considered effects to be present when the 95% credible intervals (CIs) of posterior means failed to overlap zero. Likewise, although we calculated urbanization scales of effect for all species, we focus interpretation on instances where the posterior of the urbanization coefficient had a 95% CI that did not overlap zero (Fisher et al. 2011). We analyzed models using the Program R, JAGS (Plummer 2003), and the R packages 'R2jags' (Su and Yajima 2012) and 'ncf' (Bjørnstad 2008). See Supplementary material Appendix 2 for model code.

Results

Over 34 774 total camera trap nights (N_{wint} = 14 243; N_{summ} = 20 531) we recorded 60 613 wildlife detections (N_{wint} = 24 718; N_{summ} = 35 895). The mean number of complete weeks recorded per site was 17.90 in winter (SD = 3.22) and 13.54 in summer (SD = 2.42). The most frequently detected species were white-tailed deer (N_{wint} = 6650; N_{summ} = 10 092) and raccoons (N_{wint} = 5320; N_{summ} = 9597). The least frequently detected species were red squirrels (N_{wint} = 134; N_{summ} = 235), and woodchucks (N_{wint} = 35; N_{summ} = 279).

Winter and summer models adequately fit the data (Bayesian p-values = 0.40 and 0.91, respectively). Species-specific Bayesian p-values likewise indicated acceptable fit for all species across both models (Supplementary material Appendix 4). Overall, the model assumption of spatial independence of residuals was well-met. Spline correlograms indicated a nearly complete lack of residual spatial autocorrelation (Supplementary material Appendix 3). Weak autocorrelation was only detected at short distances (< 2 km) for white-tailed deer and eastern gray squirrels in winter and red foxes in summer (Supplementary material Appendix 3). The variogram of the urbanization raster revealed a moderately fine intrinsic spatial structure, with an estimated range (and hence typical patch size) of 823 m (Fig. 3).

The magnitudes and direction of urbanization effects on mammals varied by species, season and biological response.

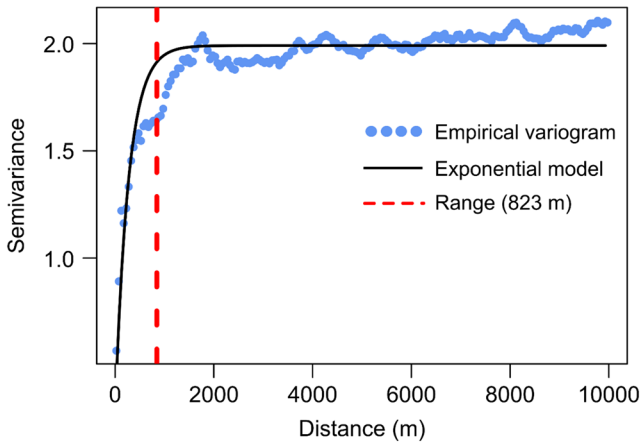

Figure 3. This variogram depicts the degree of spatial autocorrelation and intrinsic spatial structure of urbanization across Cleveland, Ohio, USA and its surrounding area (Fig. 1). The blue circles represent values from an empirical variogram; the black line is an exponential model fit to the empirical values using non-linear least squares regression; the red dashed line is the range of the variogram, indicating the distance at which spatial autocorrelation diminishes (i.e. typical patch size).

Winter occupancy was negatively affected by urbanization for eastern gray squirrels and positively affected for red foxes and Virginia opossums (Fig. 4a, Supplementary material Appendix 4). Of these, red foxes were the most strongly affected (posterior mean = 1.12; Supplementary material Appendix 4). Winter visit frequency was negatively affected by urbanization for coyotes and eastern gray squirrels and positively affected for domestic cats, fox squirrels, red foxes, white-tailed deer and woodchucks (Fig. 4b, Supplementary material Appendix 4). Of these, red foxes were the most strongly affected (posterior mean = 0.91; Supplementary material Appendix 4). Summer occupancy was negatively affected by urbanization for eastern gray squirrels and positively affected for domestic cats, red foxes and striped skunks (Fig. 4c). Of these, red foxes were the most strongly affected (posterior mean = 0.81; Supplementary material Appendix 4). Summer visit frequency was negatively affected by urbanization for coyotes, eastern gray squirrels and mustelids and positively affected for domestic cats, eastern chipmunks, eastern cottontails, raccoons, red squirrels and white-tailed deer (Supplementary material Appendix 4). Of these, red squirrels were the most strongly affected (posterior mean = 0.83; Supplementary material Appendix 4).

Scales of effect varied by species. $Scale_{90}$ was always larger than $scale_{max}$, with a mean difference of 3047 m (SD = 1217; Fig. 4, Supplementary material Appendix 4). Among the smallest scales of effect observed were domestic cat winter visit frequency ($scale_{max}$ = 19 m, 95% CI = 2–44; $scale_{90}$ = 50 m, 95% CI 10–100) and eastern cottontail summer visit frequency ($scale_{max}$ = 41 m, 95% CI = 2–127; $scale_{90}$ = 90 m, 95% CI = 10–280; Fig. 4, Supplementary material Appendix 4). Among the largest scales of effect observed were eastern chipmunk summer visit frequency ($scale_{max}$ = 6514 m, 95% CI = 2588–9809;

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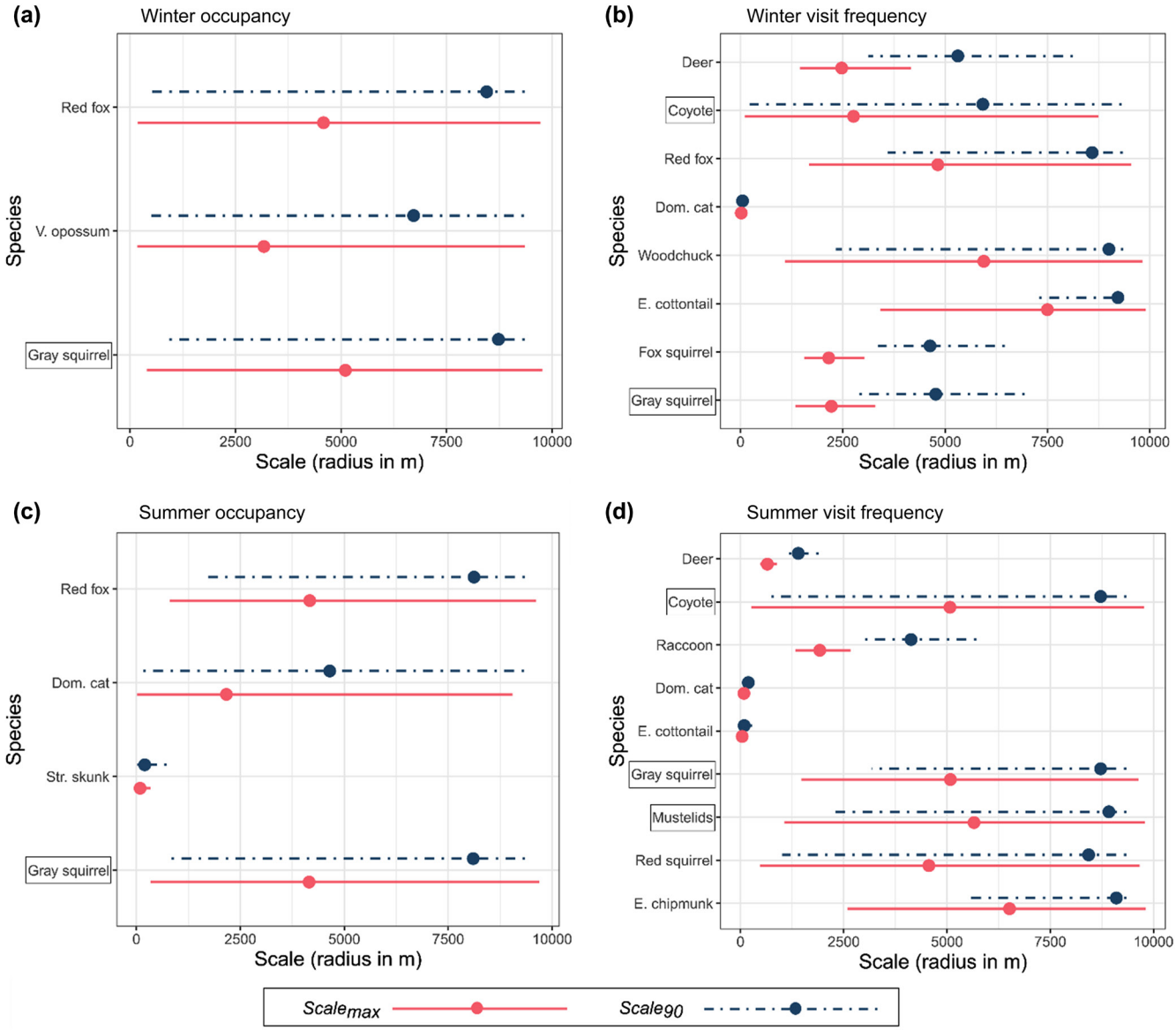

Figure 4. Scales of effect of urbanization on mammals as estimated by models fit to data from camera traps deployed from in 2015 to 2016 throughout the Cleveland Metroparks in Cleveland, Ohio, USA. Species are ordered by descending body size. Circles and lines represent posterior means and 95% credible intervals, respectively. Species names encompassed within a box indicate those which responded negatively to urbanization; all others responded positively. The maximal scale of effect ($scale_{max}$) is depicted in red and the scale containing 90% of the smoother weight ($scale_{90}$) is depicted in blue (see text for detailed regarding these scales). Winter and summer models included data December–April and June–September, respectively.

$scale_{90}$ = 9100 m, 95% CI = 5550–9350) and mustelid summer visit frequency ($scale_{max}$ = 5656 m, 95% CI = 1068–9787; $scale_{90}$ = 8920 m, 95% CI = 2300–9350; Fig. 4, Supplementary material Appendix 4). Several scales of effect estimates were accompanied by considerable uncertainty (Fig. 4). For example, the $scale_{max}$ 95% CI for red fox winter occupancy nearly spanned the entire range of scales (180–9725 m, Fig. 4a, Supplementary material Appendix 4). Similarly high uncertainty was present for Virginia opossum winter occupancy, and gray squirrel winter and summer occupancy (Fig. 4, Supplementary material Appendix 4). Conversely, the several species responses that displayed small scales of effect (i.e. $scale_{max}$ < 200 m) had precise estimates (Fig. 4, Supplementary material Appendix 4). For example, the $scale_{max}$ 95% CI for domestic cat summer visit frequency was 53–124 m (Fig. 4d, Supplementary material Appendix 4).

Scale of effect did not vary as predicted by theory for body size (positive relationship expected), mobility (positive relationship expected) or biological response (larger scale of effect expected for occupancy compared to site visit frequency). There was no evidence for a relationship between scales of effect for occupancy and species traits for either season. The

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means of the posterior means for slope parameters in these models were all near zero (ranging −0.08 from 0.04) and the associated means of their 95% CIs broadly overlapped zero ($CI_{summ_bodymass} = -0.62, 0.62$; $CI_{summ_HR} = -0.49, 0.33$; $CI_{wint_bodymass} = -0.51, 0.43$; $CI_{wint_HR} = -0.34, 0.29$). Likewise, there was no evidence for a relationship between scales of effect for visit frequency and species traits: the slopes of these relationships were all near zero, with their mean 95% credible intervals including zero (Fig. 5). There was also no evidence for differences across seasons as the grand means of posterior means and 95% CIs for season effects in these models were −0.29 (CI = −1.47, 0.90) for occupancy, and 0.01 (CI = −1.56, 1.57) for visit frequency. Finally, there was no difference in scales of effect between biological responses (occupancy versus visit frequency) pooled across seasons (i.e. occupancy versus site visit frequency), as the mean of the posterior means for this effect was −0.50 with an associated 95% CI of −0.68 to 0.98.

Discussion

The scaling of urbanization effects varied substantially across this community of 15 mammal species. The observed scales

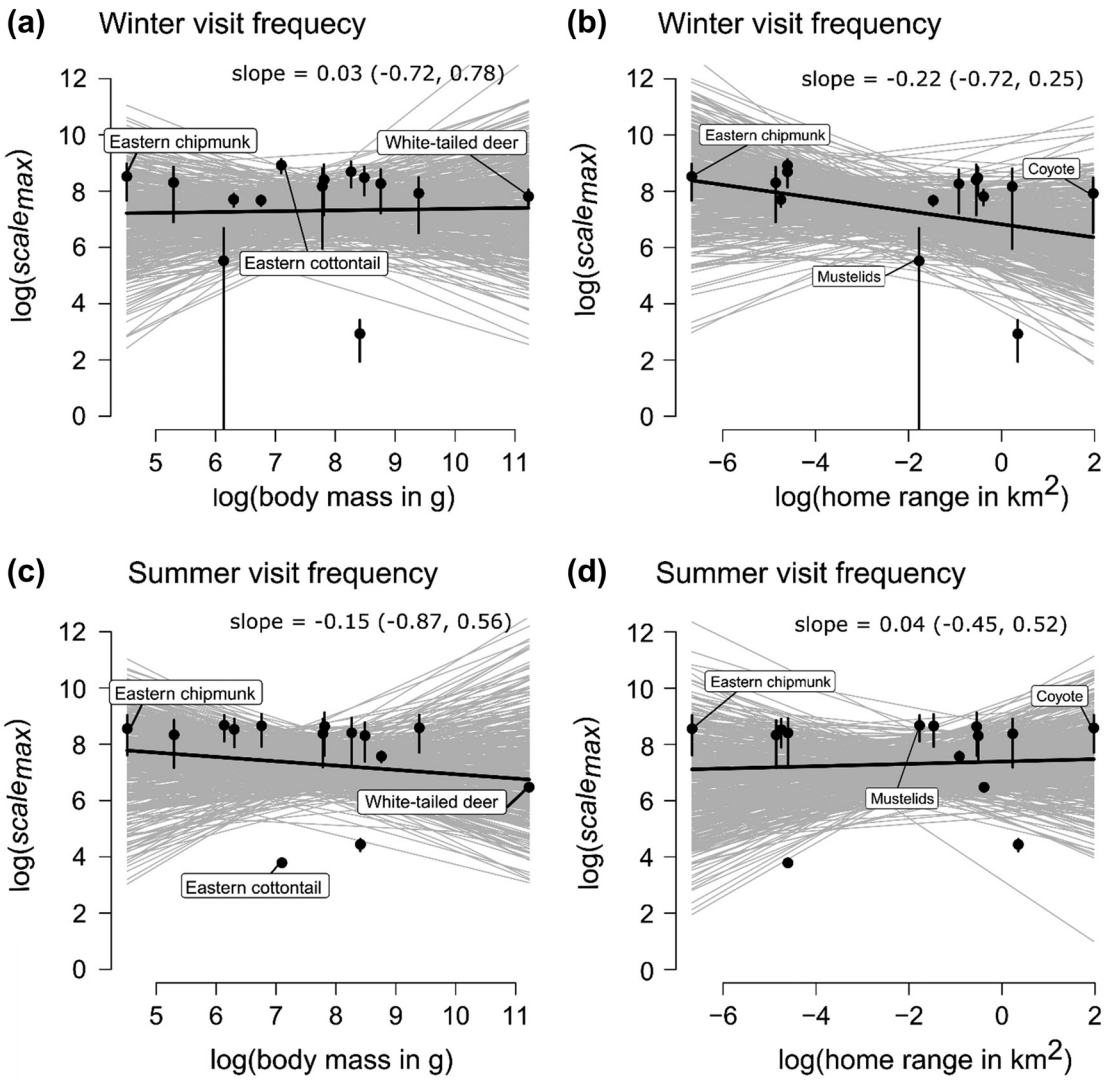

Figure 5. Plots relating scales of effect (the scale at which a maximal effect is present; $scale_{max}$) to species traits for the mammal community inhabiting the Cleveland Metroparks in Cleveland, Ohio, USA. The response variable is species' site visitation frequency during winter (December 2015–April 2016) and summer (June 2016–September 2016). Black circles and vertical lines indicate estimated scales of effect and their standard deviations estimated in the first step of analysis; the black line represents the mean predicted relationship; gray lines represent predictions of a random subset of 400 iterations from the Bayesian MCMC analysis to depict uncertainty. The slope parameters indicate the means of posterior means and 95% credible intervals of slopes estimated via a two-step Bayesian modeling process (see text for details).

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of effect (Fig. 4, Supplementary material Appendix 4) differed from the moderately fine intrinsic spatial structure of urbanization in our system (i.e. a typical patch size of ~800 m; Fig. 3), indicating these scales were not merely reflective of urban heterogeneity but rather were species-specific responses to the urban landscape (De Knegt et al. 2011). Nonetheless, these scales of effect did not vary as predicted by theory and many exhibited high uncertainty (Fig. 4, Supplementary material Appendix 4). We found no evidence of a relationship between scales of effect and species' body mass, mobility, biological response or season (Fig. 5). However, scales of effect did vary much more for species' site visit frequency compared to their occupancy (Fig. 4, Supplementary material Appendix 4). These results mirror recent studies that have found a general lack of support for theoretical predictions regarding scale of effect across several taxa (Galán-Acedo et al. 2018, Martin 2018, Gestich et al. 2019, Moraga et al. 2019), although other work has suggested that body size correlates with scale of effect in birds (Thornton and Fletcher 2014). Indeed, the spatial scaling of species–landscape relationships is an active area of research (Miguet et al. 2016, Mertes and Jetz 2018). General theory has remained elusive despite the intuitively important role scale plays in linking pattern to process for spatially-explicit ecological phenomena (Weins 1989, Levin 1992, Boyce et al. 2017). Empirical evaluations of scale have often failed to consider a wide enough range of scales to inform theory (Jackson and Fahrig 2015) and multi-species examinations of scale remain rare.

The typical approach to determining these scales involves evaluating the effects landscape covariates in concentric buffers around study sites (Miguet et al. 2016). This method, however, tends to underestimate effects at small and large scales (Aue et al. 2012) and fails to capture the tendency for local landscape features to have stronger effects on species' ecology than distant ones (Dunning et al. 1992). The smoothing-parameter approach employed here has been relatively common in stream ecology research (Comeleo et al. 1996, Peterson et al. 2011), but remains rare in studies of terrestrial species–landscape relationships (Chandler and Hepinstall-Cymerman 2016). We suggest researchers investigating scale consider this method to avoid model bias and scale underestimation (Miguet et al. 2017), especially when studying terrestrial vertebrates in landscapes with fine intrinsic spatial structure (i.e. patch sizes < 1000 m; Mertes and Jetz 2018). The smoothing approach can be used to calculate both the scale of maximal effect ($scale_{max}$) and the scale at which some larger proportion of a landscape effect is captured (e.g. 90%; $scale_{90}$). The latter is useful in situations that entail intense management, such as landscape-level decisions regarding species of conservation concern or invasive species control (cf. Miguet et al. 2017). Here, $scale_{90}$ was ~3000 m larger than $scale_{max}$ on average (Supplementary material Appendix 4; Fig. 2, 4). This disparity implies that the traditional focus on $scale_{max}$ in evaluations of scales of effect (Holland et al. 2004, Galán-Acedo et al. 2018, Gestich et al. 2019) has likely resulted in an overall underestimation of the scale of landscape effects on wildlife (Miguet et al. 2017). The smoothing approach also enables a formal estimation of scale uncertainty, which has generally been treated heuristically by applying model criteria such as AIC to find a point estimate of the scale of effect (Miguet et al. 2017, Moll et al. 2018, Moraga et al. 2019).

For most species in this study, occupancy and site visit frequency increased with increasing urbanization. This trend was evident for red foxes and domestic cats (Fig. 4). Red foxes thrive in European urban settings (Soulsbury et al. 2010), and work in North America suggests that urban areas might enable them to coexist with their dominant competitor, coyotes (Moll et al. 2018, Mueller et al. 2018). White-tailed deer site visit frequency also showed a strong positive response to urbanization (Supplementary material Appendix 4, Fig. 4b, d), which is likely related to habitat fragmentation and the associated tendency of urban deer to have smaller home ranges than their rural counterparts (Grund et al. 2002). Gray squirrel occupancy and site visit frequency had consistently negative associations with urbanization (Fig. 4). Previous research has found that, compared to fox squirrels, gray squirrels are less tolerant of patch isolation and thus the reduced connectivity induced by urbanization may limit their distribution in more urbanized areas (Zollner 2000). Overall, species' site visit frequencies were more responsive to urbanization than their occupancy (Fig. 4). More urbanized patches could be used more frequently by species for a number of reasons, including a 'piling-up' effect on species' densities due to fragmentation (Crooks et al. 2010), a concentration of anthropogenic resources (Bozek et al. 2007), or the presence of infrastructure that provides cover for rest sites or dens (Gosselink et al. 2007).

Various mechanisms have been explored in relation to the scaling of species–landscape relationships, including patch foraging behavior, perceptual abilities and dispersal capacities (Ritchie 1998, Zollner 2000, Jackson and Fahrig 2012, Miguet et al. 2016). This study provides guidance for further evaluation of such hypotheses in the context of urbanization. For example, given that eastern chipmunks perceive landscapes at a scale of ~100 m (Zollner 2000), the large scale of effect found here ($scale_{max}$ = 6514 m for summer visit frequency) suggests that a broader, and perhaps indirect effect, is at play. For instance, given that several predators hunt chipmunks in our system (e.g. red-tailed hawks *Buteo jamaicensis*), this scale of effect might relate to a broader dynamic related to variation in predation pressure over an urbanization gradient. In contrast, the small scales of effect for some species suggest that local features associated with urbanization drive their site use intensity. For example, residences likely serve as a fine-scale spatial anchors for domestic cats (Fig. 4b, d). Finally, the high uncertainty associated with some scales of effect (Fig. 4) could indicate that species' responses to urbanization are scale-independent, or that they respond to more than one urban feature on multiple scales, thus blurring specific scales. This high uncertainty was especially evident for scales of effect on occupancy, with all estimates except for one

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(striped skunk summer occupancy) exhibiting a 95% CI that nearly spanned all possible values (0–10 000 m). This finding could indicate that species' occupancy only weakly responds to urbanization in the Cleveland Metroparks, or that occupancy must be assessed at even greater extents to determine the scaling of urbanization effects (e.g. across multiple cities and expansive natural areas). We note that our study focused on parkland and did not extend to the urban core (e.g. downtown commercial districts). It is possible that the effects of urbanization on mammals and associated scales of effect could differ than the results presented here if these extremely highly urbanized sites were surveyed.

In urban systems, accurately estimating scales of effect can inform management and urban planning (Pickett et al. 2016, Moll et al. 2019a). This mammal community responded to the urban landscape at highly variable and often large scales (i.e. thousands of meters; Fig. 4). Although the mechanisms underlying these scales remain in need of examination, this variability highlights the need for increased awareness of scaling issues in urban wildlife management and landscape planning (Borgström et al. 2006). For example, urban parks vary dramatically in size, and many are smaller than the typical scale of effect found in this study (Fig. 4). Therefore, there is a need for coordinated urban landscape management across parks, which can be challenging due to the presence of multiple jurisdictions or governance bodies (Borgström et al. 2006). Ultimately, urbanization is a complex process that can be represented using a variety of variables (e.g. roads, population density, impervious surface; reviewed by Moll et al. 2019a). We represented urbanization using a principal components analysis, a method generally suited to investigate general patterns of species–landscape relationships (du Toit and Cilliers 2011, McDonnell et al. 2012). Different scales of effect might be uncovered if species were especially responsive to a specific component of the urban landscape that was obscured by a broader approach. The urbanization covariate employed here was based, in part, on forested land cover, with more urbanized locations tending to lack such cover. In urban areas lacking forest cover (e.g. cities in desert biomes), researchers might consider using an alternative non-urban land cover type when calculating a similar omnibus urbanization covariate (e.g. the 'shrub/scrub' type in the NLCD).

Although scale is often invoked as a unifying concept in ecology (Levin 1992), it has provided somewhat limited guidance for empirical work as a practical paradigm in community and landscape ecology (Miguet et al. 2016, Boyce et al. 2017, Martin 2018). This is perhaps because ecological communities are notorious 'middle-number' systems (Allen and Starr 1982) that are complex enough to be influenced by the landscape at multiple scales, but too complex to show the consistent patterns of variation that emerge at broader extents (e.g. global biodiversity gradients; Weins 1989, Kreft and Jetz 2007). Perhaps the most general result from this study is that scales of effect with respect to urbanization in this mammal community did not follow a general rule. As noted elsewhere (Miguet et al. 2016), it would be useful if scales of effects could be predicted from theory a priori because that would enable researchers or managers to plan activities without requiring resource-intensive empirical assessments of scale. Here, scales of effect lacked distinctive patterns across seasons and with respect to species traits (Fig. 4, 5). At the same time, the scales of effect related to site visit frequency were more variable, often smaller, and sometimes more precisely estimated than those for occupancy (Fig. 4, Supplementary material Appendix 4). The results of this case study suggest that body size and mobility do not closely correlate with scale of effect, and that, in this system, scales of effect are highly variable for site visit frequency while being highly uncertain and generally consistent for species' occupancy. A potentially productive way forward in building general theory for the spatial scaling of urbanization effects on mammals is for researchers to conduct studies similar to that presented here, where scales are assessed across a diversity of species and response variables, followed by aggregate-level analyses that search for general patterns (e.g. meta-analyses). Understanding the spatial scales at which wildlife respond to landscapes is crucial for both advancing ecological theory and connecting research findings to actions such as landscape management, which will suffer in effectiveness and resource efficiency if directed at scales that mismatch with species' response to landscapes (Montgomery et al. 2018).

Data availability statement

Data available from the Dryad Digital Repository: <https://doi.org/10.5061/dryad.brv15dv5j> (Moll et al. 2019b).

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Supplementary material (available online as Appendix ecog-04762 at <www.ecography.org/appendix/ecog-04762>). Appendix 1–4.

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